away from the surface. The temperature increase in the upper stratosphere and lower mesosphere produces an increase in sound speed of about 30 to 35 m/sec. Although this effect produces a "sound channel" between the surface and about 50 km, no semidiurnal or diurnal variability of the type required is known. Conceivably a rising and falling of the reflecting layer may occur from solar and auroral heating, and this may vary the positions of reflexion and produce the observed effect. Preliminary results indicate that wind variations are the more important factor.

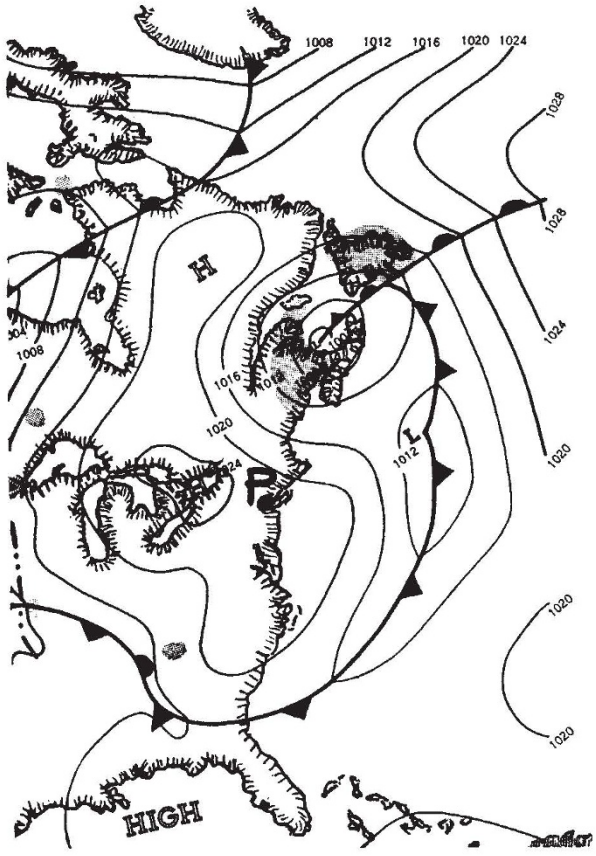


Fig. 2. Weather chart of the western North Atlantic Ocean showing the storm pattern related to the microseisms and microbaroms of Fig. 1. Palisades (Lamont Geological Observatory location) is shown at "P".

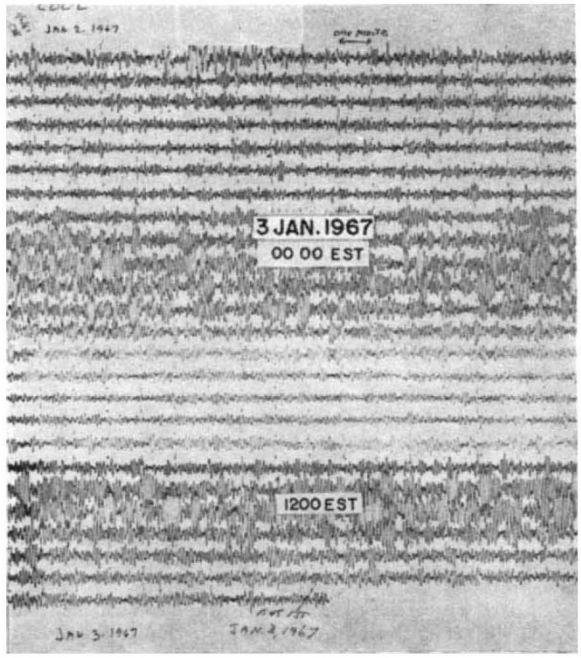


Fig. 3. A portion of the 24 h record of infrasonic waves recorded on January 2–3, 1967, at the Lamont Geological Observatory (Palisades, NY) with the use of Globe microphonic pressure transducers. About 17 min of each line, which contain 60 min, are shown in the copy. Timing marks are made at intervals of 1 min. (EST is 75° meridian time.)

Tidal wind components show a pronounced semidiurnal clockwise rotation in the northern hemisphere[7]. At an elevation of about 100 km the tidal component is reported to have a speed of about 30 m/sec. The normal track and development of cyclones are such that the most common infrasonic signal of storm origin arrives from a direction north-east to east-north-east of Palisades. According to Stolov[7], winds at about 100 km come from a north-easterly direction at about 0100–0200 and 1300–1400 local solar time. The 6 h differential in the ambient wind speed is thus 60 m/sec. The mean zonal winds above the tropopause increase from about 20 to about 70 m/sec, and so the tidal wind effect would seem, in a preliminary examination, to be of significance in producing the semidiurnal oscillation.

On at least two occasions of strong semidiurnal oscillations the signal, at times of wave amplitude maxima, arrived from well developed storms to the north-east of Palisades as determined by our tripartite station. But at the times of minima, the signals arrived from the north-west. At these times, intense storms were present off the west coast of Canada some 3,000 or more kilometres away. The tidal wind component would also be from the west at these times. The semidiurnal variation is under continuing investigation. It should be noted that this discussion refers to storm generating areas at least hundreds of kilometres from the recording station. When they are quite close to the station, microbarom activity is continuously high with no obvious time variations.

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[1] Benioff, H., and Gutenberg, B., *Bull. Amer. Meteorol. Soc.*, **20**, 421 (1939).
[2] Donn, W., and Posmentier, E., *J. Geophys. Res.*, **72**, 2053 (1967).
[3] Cook, R., *Fourth Intern. Cong. Acoustics, Copenhagen* (1962).
[4] Posmentier, E., *Geophys. J.* (in the press).
[5] Longuet-Higgins, M. S., *Phil. Trans. Roy. Soc.*, **243**, 1 (1950).
[6] Saxer, L., *Arch. Meteorol. Geophys. Bioklimatol.*, A, **6**, 451 (1954).
[7] Stolov, H., *J. Meteorol.*, **12**, 117 (1955).

Palaeomagnetism of Atlantic Islands: Fernando Noroñha

BECAUSE of their role in testing the crustal spreading hypothesis[1] the ages of Atlantic islands have assumed a prime importance[2]. Conventional geological methods are not usually applicable to studies of oceanic islands, because of the general lack of stratigraphically useful fossils and the dominantly volcanic environment. Palaeomagnetic methods, as pioneered by Einarsson[3], however, can provide valuable stratigraphic indices in volcanic regions. In addition, if defined from a time span sufficient to average out secular variation[4], palaeomagnetic poles can provide age estimates, because with increasing age they have occupied positions at increasing angular distances from the present geographic pole. Palaeomagnetic surveys are forming the bases of diverse studies of Atlantic islands. This is the third in a series of reports[5,6], in which the palaeomagnetic poles resulting from these surveys are reported.

Fernando Noroñha (Fig. 1) is an island of 18·4 km² area on the geomagnetic equator, at latitude 3° 50′ S. and longitude 32° 26′ W., approximately 345 km from the

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north-east coast of Brazil. Almeida[7] has carried out a detailed geological survey, including extensive chemical and petrological analyses, to define the volcanic history of the island. Two principal stratigraphic units are defined (Fig. 1). The older Remédios formation contains no effusive rocks, but is highly diversified, consisting of various pyroclastic rocks cut by plugs of alkali basalt, essexite porphyry, trachyte, large phonolite domes, and dykes of very variable petrological type. Above an irregular unconformity, the younger Quixaba formation consists of ankaramitic lavas and intercalated pyroclastics. The physiography is dominated by the phonolite domes, with "Pico" being regarded as "the most striking landmark in the South Atlantic Ocean"[8]. In common with outcrops on most oceanic islands, no sound age estimates are possible. Almeida limits speculation to the following: "The age is certainly pre-glacial, and considerations about the age of volcanism in the Atlantic Ocean seem to indicate that it ought not to be older than Senonian (Upper Cretaceous)"[9].

Wilson[10] noted that a pre-continental drift configuration would involve virtual superimposition of the magmatic sources of the islands of the Gulf of Guinea (Fernando Po, Principe, Annabon) and Fernando Noroñha, and that the available geochemical data are consistent with such a similarity.

At least four cores were drilled from each of twenty-eight separate bodies which represent most of the outcrops on the island. Sample locations are shown in Fig. 1. The cores, of 2·5 cm diameter and average length 12 cm, were oriented in geographic co-ordinates while still attached to outcrop. An astatic magnetometer and alternating field demagnetizing unit were used to determine the natural remanent magnetism (NRM) of each specimen. The results for each body are given in Table 1. Polished thin sections from all specimens have been examined using a Reichert 'Zetopan-pol' reflexion microscope at × 1,200 under oil, and Curie points have been measured in each specimen. Full results will be reported elsewhere, but it can be stated here that there is no reason to believe that the NRM measured is not caused dominantly by the ambient magnetic field during the initial cooling of the bodies sampled, although the NRM in four bodies is insufficiently well defined[11] to be included in group statistical analyses[12].

The virtual geomagnetic poles (or surface expressions of the hypothetical geocentric dipoles which would give rise to the observed NRM) for each body are shown in

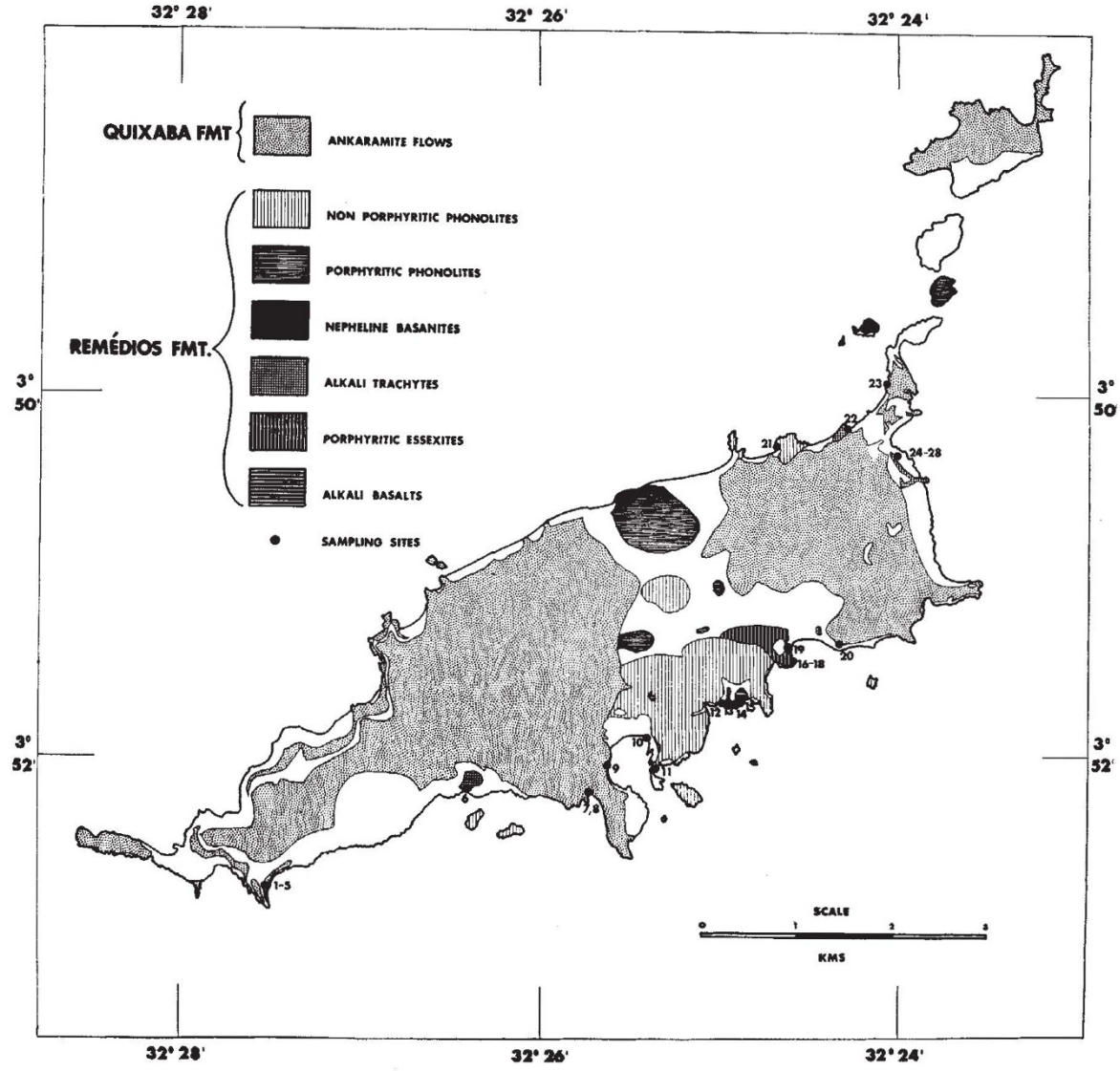


Fig. 1. Geological map of Fernando Noroñha, after Almeida[7]. Sample sites are shown as solid circles, with numbers corresponding to the data presented in Table 1.

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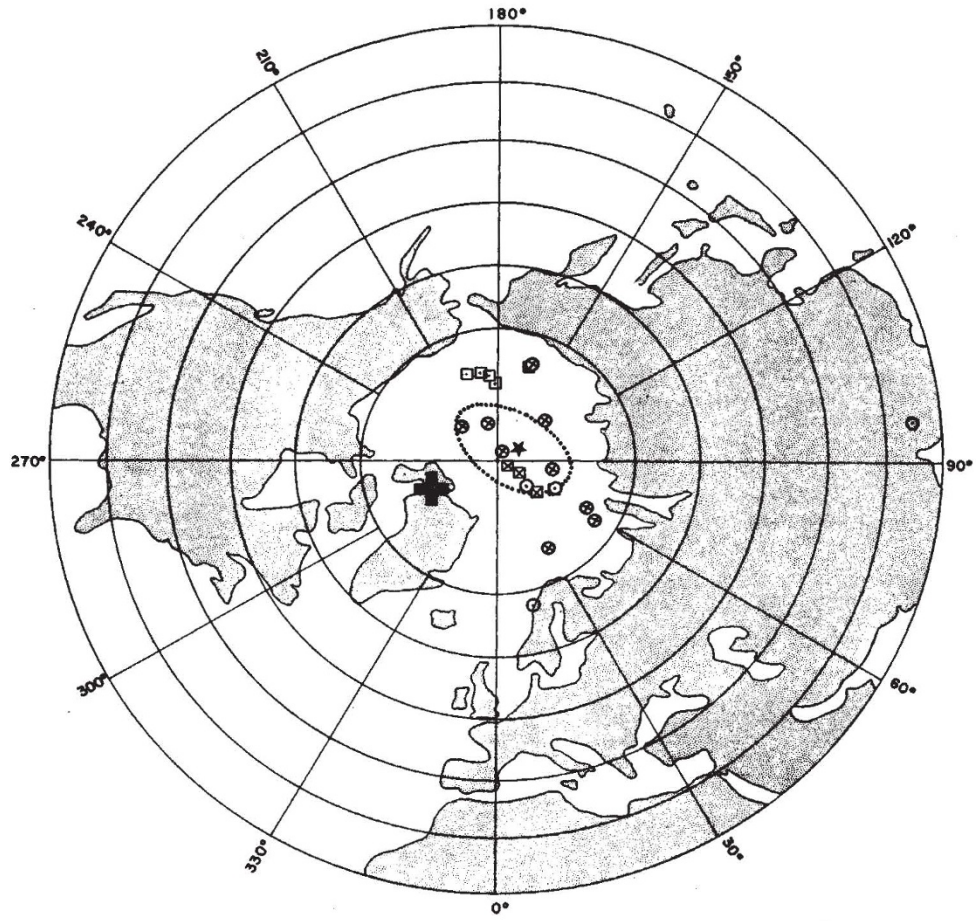


Fig. 2. Virtual geomagnetic poles for each of the twenty-four units which provide data of sufficient precision. Results from the older Remédios formation are shown as circles; those from the younger Quixaba formation are shown as squares. A cross within the circle or square signifies normal polarity; a dot within the circle or square signifies reversed polarity; (reversed or southern hemisphere poles are transposed to the northern hemisphere). The mean of all the virtual geomagnetic poles (or palaeomagnetic pole) is shown as a star. The dotted oval is the 95 per cent confidence oval of the palaeomagnetic pole. The position of the present geomagnetic pole is shown by a heavy cross.

Table 1. FERNANDO NORONHA PALAEOMAGNETIC DATA

(Each unit)

Body number	N	k_0	k_{100}	k_{200}	k_{max}	D	I	α_{95}	$\emptyset'$	θ'
					Quixaba formation					
1	4	514·6	303·5	250·3	413·7	170·5	+26·7	4·5	9·5	−76·2
2	4	364·8	491·5	393·8	857·3	171·9	+27·0	3·1	4·7	−76·9
3	4	948·2	352·8	252·0	259·1	172·6	+26·5	4·3	2·9	−77·6
4	4	89·8	258·7	166·2	417·9	173·3	+30·6	4·5	354·6	−75·8
5	4	1,033·7	436·6	581·2	876·5	169·0	+23·8	3·1	18·9	−76·2
7	4	35·5	195·1	30·0	195·1	358·7	−06·8	6·6	78·7	88·8
8	4	42·5	29·4	19·5	30·7	356·5	−06·8	16·8	65·2	86·5
9	4	101·7	128·6	149·6	151·4	350·6	−04·8	7·5	66·3	80·5
20*	4	0·9	0·9	0·9	1·0					
23	4	3·9	133·3	154·3	214·9	352·8	−08·1	6·3	55·9	82·9
					Remédios formation					
6	4	4·3	6·0	2·0	6·1	355·7	−33·3	40·7	163·8	75·1
10	4	15·4	33·3	24·8	36·3	14·2	−09·5	15·4	61·7	75·9
11	4	11·4	3·7	22·8	25·8	53·0	+77·2	18·4	347·3	10·8
12	4	80·7	122·3	236·3	239·7	189·1	+10·2	5·9	246·6	−80·8
13*	4	1·9	2·1	2·4	2·8					
14	4	20·5	70·5	72·0	83·6	3·8	−00·1	10·1	192·1	84·6
15	4	1·9	3·2	7·1	38·0	196·5	−23·0	15·1	193·3	−67·2
16*	4	1·4	1·4	1·6	1·6					
17	4	139·2	431·0	320·7	604·4	355·9	−33·9	3·7	162·5	74·8
18	4	50·6	693·6	582·3	779·2	185·3	+05·0	3·3	223·0	−84·6
19	4	51·4	173·1	13·9	173·1	359·7	−09·9	7·0	165·3	88·9
21	5	2·1	7·8	1·0	9·7	240·3	+56·0	25·9	276·6	−25·9
22	4	17·5	60·5	39·6	60·5	353·6	−01·3	11·9	81·4	81·9
24	4	44·9	116·9	288·9	304·4	13·0	−05·9	5·3	30·0	75·3
25	4	28·5	33·3	10·5	33·3	7·6	−05·0	16·1	227·3	82·2
26*	4	3·6	3·0	2·5	3·1					
27	4	5·6	27·6	9·2	47·0	343·9	−06·8	13·5	58·9	73·9
28	4	9·1	46·7	1·7	46·7	1·8	−24·5	13·6	136·5	80·7

N is the number of cores. k_0, k_{100}, k_{200} is the Fisher[12] precision parameter of original NRM, and following demagnetization in alternating magnetic fields of 100 and 200 oersteds respectively. k_{max} is the maximum precision parameter obtained by combining either 100 or 200 oersted NRM directions to obtain minimum scatter. D and I are the declination and inclination of NRM, east of geographic north, and with respect to horizontal (+ is below, − is above). α_{95} is the semi-vertical angle of 95 per cent confidence cone. $\emptyset'$ and θ' are longitude and latitude of virtual geomagnetic poles, in degrees east, with negative sign indicating southern latitudes. D, I, α_{95}, $\emptyset'$, and θ' are all based on k_{max} NRM, with unit vector per core.

* Indicates that the data do not provide an NRM direction with sufficient precision[11], and hence data are rejected.

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Table 2. FERNANDO NOROÑHA PALAEOMAGNETIC DATA

(Group analysis)

Group	B	N	k_0	k_{100}	k_{200}	k_{max}	D	I	a_{95}	$\emptyset'$	θ'
Quixaba fmt.	9	36	29·6	57·0	49·7	52·1	353·0	−17·9	7·2	20·1	81·2
Remédios fmt.	15	61	6·8	6·6	5·2	6·5	6·0	−07·6	16·2	237·2	84·0
All units	24	97	9·6	8·9	7·5	9·1	0·8	−11·9	10·4	126·5	87·7

Symbols as in Table 1, except here unit vector per body is applied. In addition B is the number of bodies sampled. Note that in some of the above cases $k_{max} < k_{100}$. This is not a computational error: k_{max} results from unit vector being applied to each 'least scatter' (k_{max}) body as in Table 1, and there is no reason why this must be $> k_{100}$ or k_{200}.

Fig. 2. Table 2 shows the results of combining the data to obtain the palaeomagnetic poles for both formations, and for the entire survey. The palaeomagnetic pole for the entire survey is shown in Fig. 2, together with the 95 per cent confidence oval. Polarity variations in both formations suggest a sufficient time span for probable cancellation of short period or secular variations[13]. Fig. 2 clearly shows that the palaeomagnetic pole for Fernando Noroñha is not significantly different from that caused by the present axial dipole. Comparison of this fact with much independent data strongly suggests that Fernando Noroñha does not exceed Miocene age in outcrop. For example, Doell and Cox[14] suggest that post-Miocene geomagnetic poles are not on average divergent for the present geographic pole; and Irving[15] shows a clear difference between Lower Tertiary palaeomagnetic poles and the present geographic pole.

This suggested age limitation for Fernando Noroñha would clearly invalidate any comparison of the island surface with the islands of the Gulf of Guinea in terms of any pre-drift configuration, for even if continental drift is considered proved the Gulf of Guinea and north-eastern Brazil could not be envisaged as being sufficiently close to yield the same magmatic sources for outcrops of the respective islands during the Upper Tertiary.

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[1] Hess, H. H., *Geol. Soc. Amer.*, Buddington volume, 599 (1962).
[2] Wilson, J. T., *Nature*, **198**, 925 (1963).
[3] Einarsson, T., *Adv. Phys.*, **6**, 238 (1957).
[4] Cox, A., and Doell, R. R., *Bull. Seis. Soc. Amer.*, **54**, 2243 (1964).
[5] Watkins, N. D., Richardson, A., and Mason, R. G., *Earth Planet. Sci. Lett.*, **1**, 225 (1966).
[6] Watkins, N. D., Richardson, A., and Mason, R. G., *Earth. Planet. Sci. Lett.*, **1**, 471 (1966).
[7] Almeida, F. F. M., in Monog. XIII, *Div. Geol. e Min., Dep. Nac. Prod. Min. Minist. Agril., Serv. Graf. Inst. Brasil de Geog. Estat. Rio de Janeiro* (1958).
[8] Branner, J. C., *Amer. J. Sci.*, **37**, 152 (1889).
[9] Almeida, F. F. M., ref. 7, page 177.
[10] Wilson, J. T., *Phil. Trans. Roy. Soc.*, **258**, 156 (1965).
[11] Vincenz, S. A., and Bruckshaw, J. McG., *Proc. Camb. Phil. Soc.*, **56**, 21 (1960).
[12] Fisher, R. A., *Proc. Roy. Soc.*, **217**, 295 (1953).
[13] Runcorn, S. K., *Ann. Geophys.*, **15**, 87 (1959).
[14] Doell, R. R., and Cox, A., *Adv. Geophys.*, **8**, 294 (1961).
[15] Irving, E., in *Palaeomagnetism and its Applications to Geological and Geophysical Problems*, 124 (John Wiley and Sons, New York, 1964).

Additional Theory of Origin of Fiamme in Ignimbrites

THE pantelleritic Quaternary volcano Fantale (lat. 8° 58′ N., long. 39° 54′ E.) is situated at the northern end of the main Ethiopian rift valley. During a field excursion to this volcano, we made a brief examination of a recent ignimbrite. This pale green ash flow, which was probably erupted in association with the formation of the Fantale Caldera[1], is particularly rich in fiamme. Fiamme are flattened, black, glassy inclusions with flame-like cross sections. These are usually considered to have formed by the collapse of included molten pumice under the load of the overlying portions of the pyroclastic flow[2]. As we shall show, it seems unlikely that the fiamme in the Fantale ignimbrite originated in this way; the purpose of this communication is to suggest an alternative origin.

The Fantale ignimbrite covers much of the plain of Matahara surrounding the volcano to the south and west, and is also present in places on the slopes of the mountain. As the unit is traced uphill towards the caldera, however, the ignimbrite becomes much less extensive. The thickness varies considerably because of the irregular topography of the volcano before the eruption. Nevertheless, it can be shown that the unit is thinner (0·3–2 m) on the sloping flanks of the volcano, and very much thicker on the surrounding Matahara Plain (about 20 m). A striking feature is the absence of an air-fall pumice deposit either below or above the welded tuff. Wherever erosion or faulting allows examination of its whole thickness, the ignimbrite is revealed as being completely welded from top to bottom, in the more extreme cases to a hard flinty rock.

Fiamme are present throughout the flow, and although not regularly distributed they are definitely not localized in the lower parts of the unit. Even when the ignimbrite is only 30 cm thick, well developed, severely flattened fiamme are present. Under the microscope these show no evidence of a collapsed pumice origin, but instead many are filled with spherical bubbles. It is striking that the highly flattened inclusions contain generally undeformed vesicles. Similar fiamme are also present in the topmost part of the flow on the Matahara Plain. In the absence of any thick overlying tuff therefore it is unlikely that the fiamme in many parts of this unit could have originated by the collapse of included molten pumice. This is particularly obvious because the presence of euhedral uncorroded anorthoclase phenocrysts suggests that unusually high eruptive temperatures were not involved.

It is suggested that on eruption the Fantale ignimbrite was composed of the usual gas phase loaded with shards, crystals and lumps of molten pumice. In addition it contained abundant inclusions of non-vesiculated, gas-rich molten glass. These glassy inclusions must have been very fluid on eruption, probably because of their particularly high gas content[3]. Thus on deposition they became very flattened, even when the weight of the overlying ignimbrite was very small. After deposition and flattening, the inclusions at last began to vesiculate[4], forming the spherical bubbles.

It should be stressed that we do not suggest that fiamme in "normal" ignimbrites originated in this way. The mode of origin outlined here, however, must be considered when fiamme are found in extremely thin ignimbritic units.

Further work on this striking unit is in progress. Studies of present day ignimbrites in Alaska[5] indicate that "normal" ash flow deposits do not show the extreme welding and flattening in the absence of any overload, as is visible at Fantale. Particular efforts are being made to understand this feature. Other workers interested in this problem are invited to write to us; we shall be pleased

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